

SENSORS For ROBOTICS And ARTIFICIAL INTELLIGENCE

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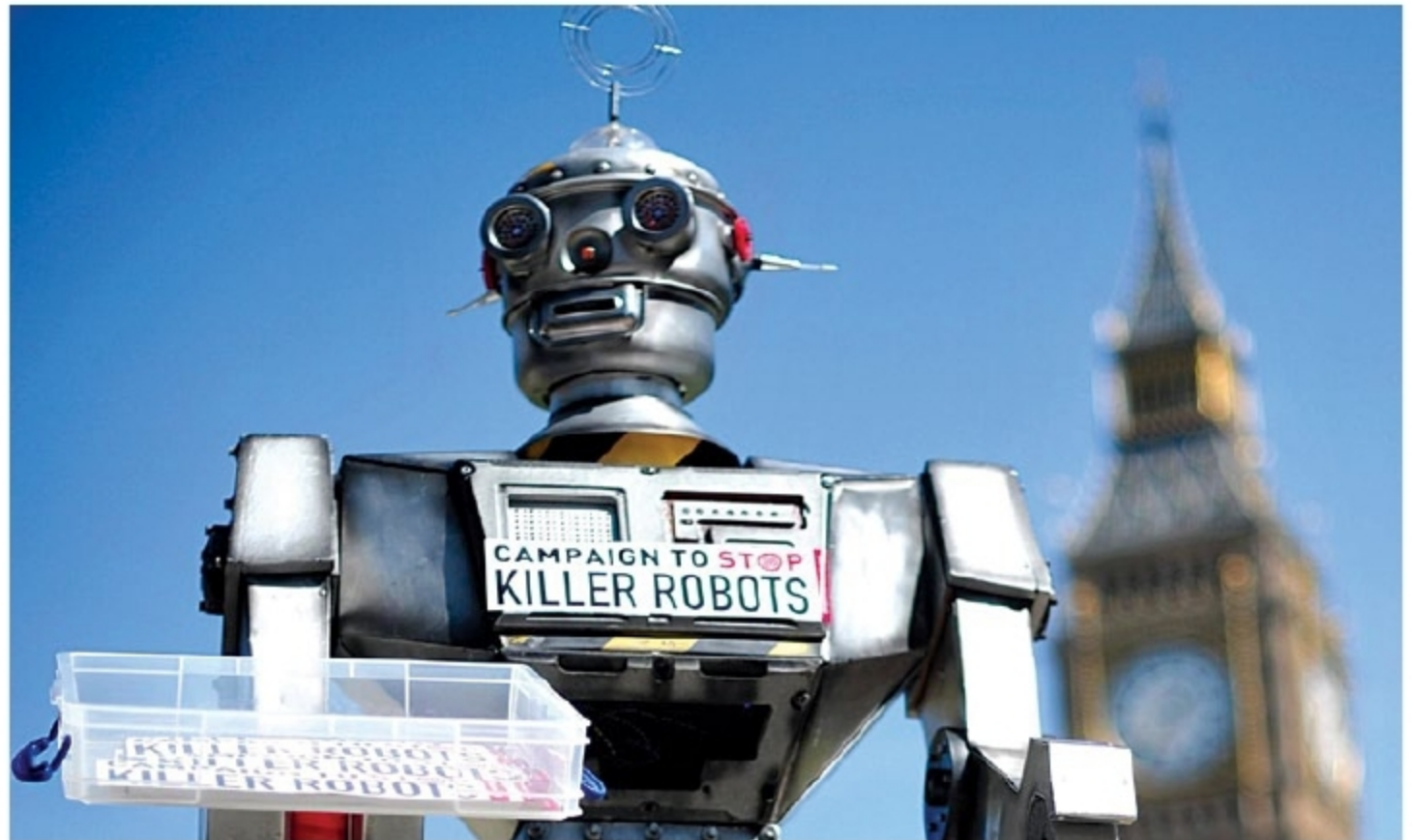


Fig. 1: Campaign to stop killer robots (Credit: www.theguardian.com)

New and advanced technologies have recreated most human sense organs, or sensors, in robots including eyes, ears, skin and nose. Just like sense organs play an important role in the human body, electronic sensors play a vital role in robotics and artificial intelligence (AI).

Robotics and AI serve different purposes. Robotics involves building robots, whereas AI deals with programming intelligence or

making computers behave like humans. Accordingly, the context of sensors in robotics and AI may be slightly different. Robots are programmable machines, interact with the physical world via sensors and actuators, and can be made to work autonomously or semi-autonomously. However, many robots have limited functionalities and are not artificially intelligent. Some industrial robots can only be programmed to perform a repetitive task, which does not require AI. Some AI algorithms are often required to allow the robots to perform more complex tasks.

Human-like operations of machines induced by AI include speech recognition, planning, problem solving and learning, among other tasks. Not all AI algorithms are used to control robots. For example, AI algorithms are used in virtual assistants like Google Assistant, Amazon Alexa and Apple Siri.

When robotics and AI are combined, we get artificially-intelligent robots. However, without sensors, AI robots are blind, deaf and dumb. Machine perception, learning,

Fig. 2: Robotic arm used while shooting Gravity, a science fiction thriller (Credit: www.derivative.ca)

